

Design of High Speed Transimpedance Amplifier for Optical Communication Systems

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Abstract

This paper presents a low noise and high bandwidth transimpedance amplifier (TIA) for optical receiver, operating at data rate of 10 Gb/s. The first configuration of the amplifier which is the basic structure is based on push-pull inverter with resistor in feedback. However, the second configuration employs an inductor in series with a resistor in the feedback to extend the bandwidth and reduce the effect of parasitic capacitances. To optimize the input referred noise of the final design, an active device is used to replace the feedback resistor. Using 0.18 μm CMOS technology, simulation results for optimized structure show a bandwidth of 7.9 GHz, transimpedance gain of 50.8 dB Ω , and input referred noise current of 7.4 pA/ $\sqrt{\text{Hz}}$.

keywords—Optical receiver; transimpedance; bandwidth enhancement ; inductor peaking.

1 Introduction

The rapid expansion of communication services and need for high data rate have undoubtedly pushed the demand for low cost systems operating at very high frequencies [1]. Consequently, optical transmission systems based on optical fibre are highly recommended to achieve the goal for short and long distances. The optical fibre is used for its advantages such as the insensitivity to spurious noise signals, and ability to avoid electrical ground loops [2]. For cost saving purpose, CMOS technology presents a good candidate to design low cost and low power TIA for optical communication systems [1]. Furthermore, the optical transmission systems require specific high transmission speeds as defined by synchronous optical network SONET and synchronous digital hierarchy SDH, the standards OC48, OC192, and OC768

corresponding to data rate of 2.5, 10, and 40 Gb/s, respectively [3] [4].

A high speed optical receiver generally consists of photodiode, transimpedance amplifier (TIA), and limiting amplifier (LA). The photodiode transforms optical received waves to small currents which are also converted and amplified to voltage signals by TIA [2] [5]. However, to support high data rate applications, widening bandwidth is the most important issue of the receiver and the TIA in the particular. Actually, the TIA is the most critical building block in the optical receiver for its contradictory requirements such as high gain, large bandwidth, and low input noise. To satisfy TIA requirements, three famous configurations are mainly developed in the literature [6]. The two first ones are based in open loop topology. However, the difference between them is in high or low input impedance as shown in the Fig.1.(a). Besides, the third configuration is closed loop based topology that uses a classical approach to cover the other topologies problems. Its main advantage is the possibility to achieve a good trade-off between dynamic range, bandwidth, gain, and input noise.

As mentioned in the literature, to achieve a high data rate, several bandwidth extension techniques can be used like feedback configuration [7], capacitive peaking [8], common-gate configuration, and the popular inductive peaking [4] [8]. Mounted in different ways, the inductive peaking technique often appears in the circuit as shunt peaking inductor, multiple inductive series peaking, π type inductor peaking (PIP), and inductive feedback [1], [9–12].

This paper is organized as follows: Section II presents a topology for this TIA. Section III gives more details about the TIA design with inductive feedback technique. Simulation and results are given in the last section. Finally, the presented work is ended by conclusions.

2 Transimpedance amplifier topology

Figure.1 presents both open and closed loop topologies of the amplifier for high-speed optical receivers.

where C_i represents the total input capacitance of the amplifier, R and R_f are the input and feedback resistances, respectively. The open loop topology, with a low input resistance, the configuration gives a large bandwidth with low gain and high level of noise. Besides, the configuration with

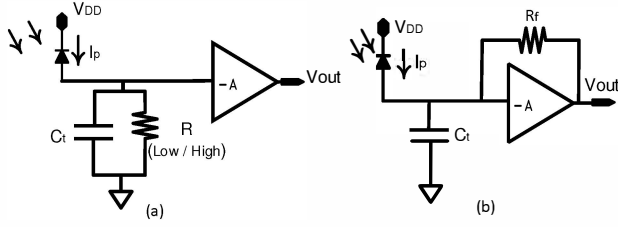


Figure 1: TIA topologies : (a) Open loop , (b) Closed loop

a high input resistance gives a narrow bandwidth with good gain and level of noise. Therefore, for trade-off aim, the open loop topology is replaced by closed loop topology using a feedback resistance that can be lowered by the voltage gain "A". Consequently, the bandwidth is extended with good gain and acceptable level of noise.

The TIA gain is defined as a ratio of its output voltage to the input current of the photodiode. According to closed loop topology, the TIA gain is given by Eq. 1. From this equation, the gain mainly depends on the feedback resistance R_f . To increase the gain, R_f must be large without affecting the stability of receiver.

$$\frac{V_{out}}{I_{in}} = -\frac{A}{(A+1)} \frac{R_f}{1 + \frac{R_f C_d}{A+1} s} \quad (1)$$

The upper frequency f_{-3dB} defines the TIA bandwidth, at which the gain dropped 3 dB below its DC value, and it is directly related to the open loop voltage gain as given in following equation :

$$f_{-3dB} = \frac{1+A}{2\pi R_f C_i} \quad (2)$$

From Eq.2, the dominant pole is defined by the time constant $R_f C_i$ that limits the bandwidth of the system. Therefore, the choice of photodiode allows to increase the bandwidth if its capacitance C_{pd} is low.

The TIA negative feedback is used to obtain stable gain and large bandwidth [2]. On other hand, the design of the system with negative feedback could cause stability problem. Therefore, to avoid this problem, a trade-off is necessary between gain and bandwidth. For high frequency applications, while the dominant pole of the open loop frequency response is located at the input node because the total input capacitance C_i , the remain poles of the amplifier must be located above the cut-off frequency as non-dominant poles to ensure a good stability. [13].

3 TIA design

3.1 Basic structure

To increase the gain bandwidth product, the proposed structure of the TIA is based on the push-pull inverter presented

by Fig.2 to maximize the amplifier transconductance [6]. Based on the small signal model of the MOSFET, the open gain voltage of the push-pull inverter is given by the following equation :

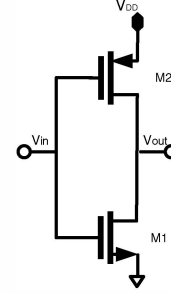


Figure 2: Push-pull inverter

$$A_{OpenLoop} = -\frac{g_{m1} + g_{m2}}{g_{ds1} + g_{ds2}} \quad (3)$$

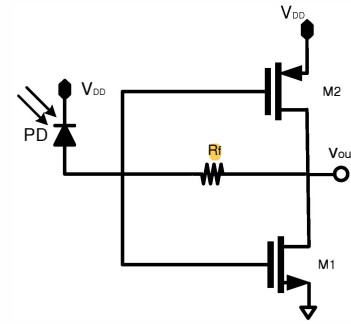


Figure 3: Basic TIA structure with R_f feedback

The transfer function of the structure depicted by Fig.3 is calculated as follows:

$$\frac{V_{out}}{I_{in}} = \frac{X(s)}{Y(s)} \quad (4)$$

where

$$X(s) = R_f C_f s + (1 - R_f g_m) \quad (5)$$

and

$$Y(s) = [R_f (C_i C_f + C_i C_o + C_f C_o)] s^2 + [(C_i + C_o) + R_f (g_o (C_f + C_i) + g_m C_f)] s + [(g_{ds} + g_m)] \quad (6)$$

where $g_m = g_{m1} + g_{m2}$ and $g_{ds} = g_{ds1} + g_{ds2}$ correspond respectively to the total transconductance and conductance of push-pull inverter. The total capacitances at input, output and the feedback of the TIA with the photodiode are represented by $C_i = C_{gs1} + C_{gs2} + C_{PD}$, $C_o = C_{db1} + C_{db2}$, and $C_f = C_{gd1} + C_{gd2}$, respectively.

3.2 Bandwidth improvement

To improve the bandwidth while reducing the input referred noise, the inductive feedback technique is used. By adding the inductor in the feedback as depicted in Fig.4, the new transfer function of system **adds one zero and one pole** compared to the first transfer function given in Eq.4,7 and 8

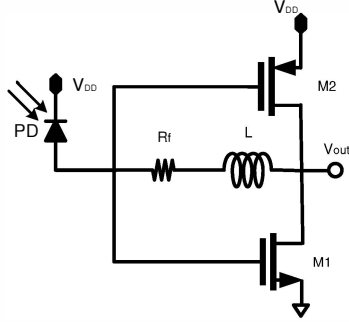


Figure 4: TIA with R_f and L in feedback

where

$$X(s) = LC_f s^2 + (RC_f - Lg_m)s + (1 - Rg_m) \quad (7)$$

and

$$Y(s) = L(C_i C_f + C_i C_o + C_f C_o)s^3 + [R(C_i C_f + C_i C_o + C_f C_o) + L(g_{ds}(C_i + C_f) + C_f g_m)]s^2 + [(C_i + C_o) + R(g_{ds}(C_f + C_i) + g_m C_f)]s + [(g_{ds} + g_m)] \quad (8)$$

As the new transfer function is **a third-order based system, it is computationally difficult to calculate the optimized value of L** . To overcome this issue, graphical or numerical methods can be used.

3.3 Noise analysis

The **input-referred current noise is one of the most critical** TIA parameters. Often the noise of the TIA dominates all other noise sources and therefore determines the sensitivity of the receiver. The equivalent noise power density at the input is given by Eq.9 [14].

$$\overline{I_{tot}^2} = \frac{4KT}{R_{eq}}B + \frac{8KT}{3g_m} \left[\frac{(1+A)^2}{R_{eq}^2}B + (2\pi C_t)^2 B^2 \right] \quad (9)$$

Where R_{eq} is the feedback resistance including R_f and the series resistance of the inductor, K is Boltzmanns constant, T is the absolute temperature, C_t is the total input capacitance and B indicates the useful bandwidth.

In order to optimize the noise, the feedback resistor R_f of the TIA is replaced by *MOSFET* device biased in the triode

region where its value can be adjusted via the gate voltage V_G as depicted in the Fig.5. The value of the *MOSFET* resistance is given by the Eq.10.

$$R_f = \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_T)} \quad (10)$$

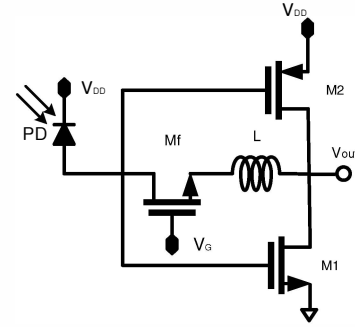


Figure 5: TIA with MOSFET and L in feedback

4 Simulation and results

The proposed TIA is designed and simulated using **0.18 μm CMOS technology** with supply voltages of 1.8 V. Based on parametric analysis, the simulation in Fig.6 shows TIA frequency responses with L variation where the optimum value of L is about 7 nH. In the depicted frequency response of

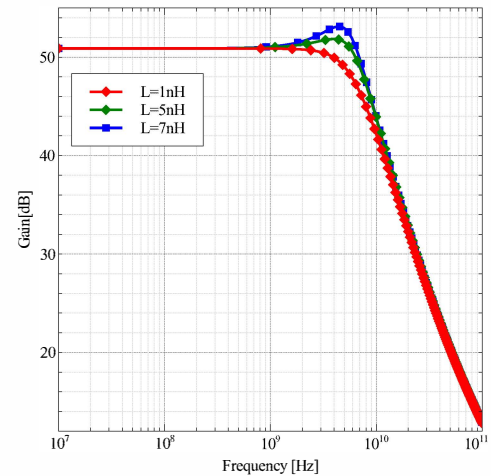


Figure 6: Frequency response of the TIA with inductor effect

Fig. 7, there are two curves with a same DC gain of 50.8 dB. With simple configuration of Fig.3, the bandwidth achieves 5.6 GHz. However, the bandwidth increases to 7.9GHz when the inductive feedback is used (see Fig.4) corresponding to a bit rate of 10 Gb/s of the SONET standard.

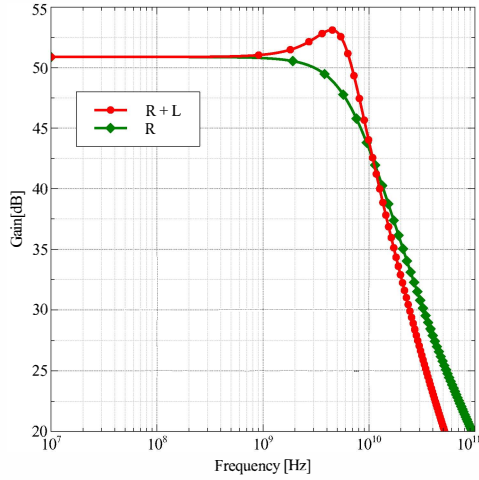


Figure 7: Frequency response of the TIA

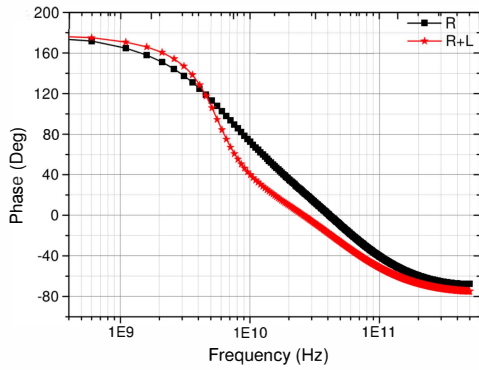


Figure 8: Phase response of TIA

To study the stability of the system, Fig.8 presents the phase of the amplifiers versus frequency. The phase margin is about 117° , 100° for the simple and inductive peaking structures, respectively. According to the Bode theorem, one can conclude that the structure with **inductive peaking technique offers more stability than the basic one without L**.

Figure.9 shows the input referred noise current for the three configurations Fig.3, 4, and 5. In low frequencies the two structures depicted in Fig.3 and 4 have the same level of noise that increases above 1 GHz . In high frequencies, the input referred current noise of the simple structure is about $11.4\text{ pA}/\sqrt{\text{Hz}}$ at cut-off frequency. When the inductive feedback is used this level decreases to $7.82\text{ pA}/\sqrt{\text{Hz}}$. Using MOSFET transistor, the noise is improved (Fig.9) where the noise level is about $6.3\text{ pA}/\sqrt{\text{Hz}}$ in low frequencies and $7.7\text{ pA}/\sqrt{\text{Hz}}$ at the cut-off frequency.

Using DC analysis, Fig.10 illustrates the output voltage versus the photocurrent. The obtained characteristic shows a good linear evolution as well as a dynamic range until 2 mA .

Table.1 summaries simulated parameters of the three TIA

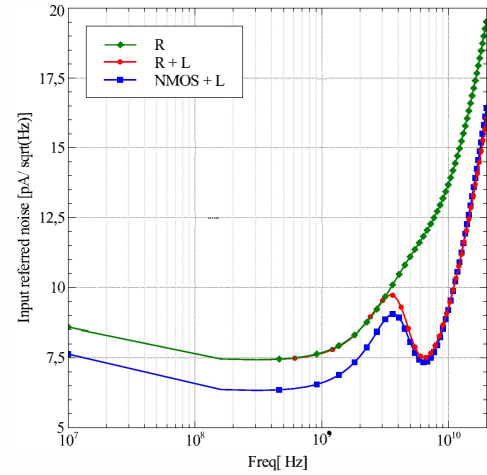


Figure 9: Input referred noise of TIA

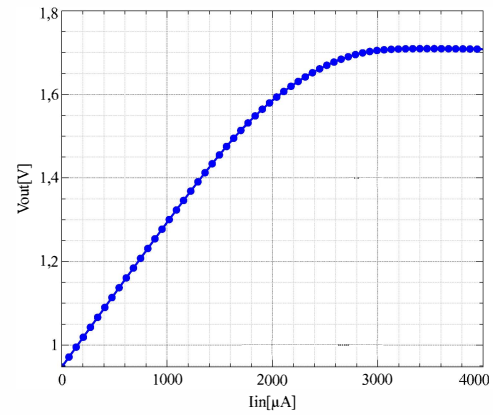


Figure 10: Output voltage vs the input photo-current

structures. The circuit performances of third the proposed TIA are summarized in Table 4 together with the state-of-the-art TIAs using a similar technology $0.18\text{ }\mu\text{m}$.

Parameters	<i>R</i>	<i>R + L</i>	<i>MOSFET + L</i>
TIA gain [dBΩ]	50.8	50.8	50.8
Bandwidth [GHz]	5.6	7.9	7.9
Noise* [$\text{pA}/\sqrt{\text{Hz}}$]	11.4	7.8	7.7

* Input Referred Noise

Table 1: Performances of the three TIA configurations

Ref	[15]	[12]	This Work
Freq [GHz]	7	7.2	7.9
Gain [dBΩ]	55	61	50.8
Input Current Noise [$\text{pA}/\sqrt{\text{Hz}}$]	17.8	8.2	7.7
P_{DC} [mW]	18.6	7.7	7.2
CMOS Technology [μm]	0.18	0.18	0.18

Table 2: Performances comparison with other TIA works

5 Conclusion

The proposed transimpedance amplifier combines push-pull inverter with peaking inductive technique. Using $0.18\mu\text{m}$ CMOS technology, the proposed design presents good performances in terms of bandwidth, input referred noise current and gain. One can conclude that the introduction of inductive peaking in the feedback has improved the bandwidth, and thus, the data rate.

References

- [1] J.-D. Jin and S. Hsu, "A 40-Gb/s transimpedance amplifier in $0.18 - \mu\text{m}$ cmos technology," *Solid-State Circuits, IEEE Journal of*, vol. 43, pp. 1449–1457, June 2008.
- [2] C. Hermans and M. Steyaert, *Broadband Opto-Electrical Receivers in Standard CMOS*. Analog Circuits and Signal Processing Series, Springer Netherlands, 2007.
- [3] J. Cao, M. Green, A. Momtaz, K. Vakilian, D. Chung, K.-C. Jen, M. Caresosa, X. Wang, W.-G. Tan, Y. Cai, I. Fujimori, and A. Hairapetian, "Oc-192 transmitter and receiver in standard $0.18 - \mu\text{m}$ cmos," *Solid-State Circuits, IEEE Journal of*, vol. 37, pp. 1768–1780, Dec 2002.
- [4] I. Song, *Multi-Gbit/s Transimpedance Amplifier with Integrated Photodetector for Optional Interconnects*. PhD thesis, School of Electrical and Computer Engineering, Georgia Institute of Technology, November 2004.
- [5] B. Razavi, *Design of Integrated Circuits for Optical Communications*. New York, NY, USA: McGraw-Hill, Inc., 1 ed., 2003.
- [6] M. Ingels, G. Van der Plas, J. Crols, and M. Steyaert, "A cmos 18 thz omega; 248 mb/s transimpedance amplifier and 155 mb/s led-driver for low cost optical fiber links," *Solid-State Circuits, IEEE Journal of*, vol. 29, pp. 1552–1559, Dec 1994.
- [7] M. Atef, "Transimpedance amplifier with a compression stage for wide dynamic range optical applications," *Microelectronics Journal*, vol. 46, no. 7, pp. 593 – 597, 2015.
- [8] F.-T. Chien and Y.-J. Chan, "Bandwidth enhancement of transimpedance amplifier by a capacitive-peaking design," *Solid-State Circuits, IEEE Journal of*, vol. 34, pp. 1167–1170, Aug 1999.
- [9] B. Analui and A. Hajimiri, "Bandwidth enhancement for transimpedance amplifiers," *Solid-State Circuits, IEEE Journal of*, vol. 39, pp. 1263–1270, Aug 2004.
- [10] H. Escid, S. Salhi, and A. Slimane, "Bandwidth enhancement for $0.18\mu\text{m}$ cmos transimpedance amplifier circuit," in *Microelectronics (ICM), 2013 25th International Conference on*, pp. 1–4, Dec 2013.
- [11] O. Ghasemi, R. Raut, and G. Cowan, "A Low Power Transimpedance Amplifier Using Inductive Feedback approach in 90nm CMOS," in *ISCAS: 2009 IEEE International Symposium on Circuits and Systems*, vol. 1, pp. 1937–1940, 2009.
- [12] C.-H. Wu, C.-H. Lee, W.-S. Chen, and S.-I. Liu, "Cmos wideband amplifiers using multiple inductive-series peaking technique," *Solid-State Circuits, IEEE Journal of*, vol. 40, pp. 548–552, Feb 2005.
- [13] M. Ingels and M. Steyaert, "A 1-gb/s, 0.7- μm cmos optical receiver with full rail-to-rail output swing," *Solid-State Circuits, IEEE Journal of*, vol. 34, pp. 971–977, Jul 1999.
- [14] R. Smith and S. Personick, "Receiver design for optical fiber communication systems," in *Semiconductor Devices for Optical Communication* (H. Kressel, ed.), vol. 39 of *Topics in Applied Physics*, pp. 89–160, Springer Berlin Heidelberg, 1982.
- [15] Z. Lu, K. S. Yeo, W. M. Lim, M. A. Do, and C. C. Boon, "Design of a cmos broadband transimpedance amplifier with active feedback," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 18, pp. 461–472, March 2010.